INFRASTRUCTURE

THE BEHEMOTH

How five years, 964 piles, 14 miles of cable, and a few sturgeon come together to build a record-setting bridge. **BY DAVEY ALBA**

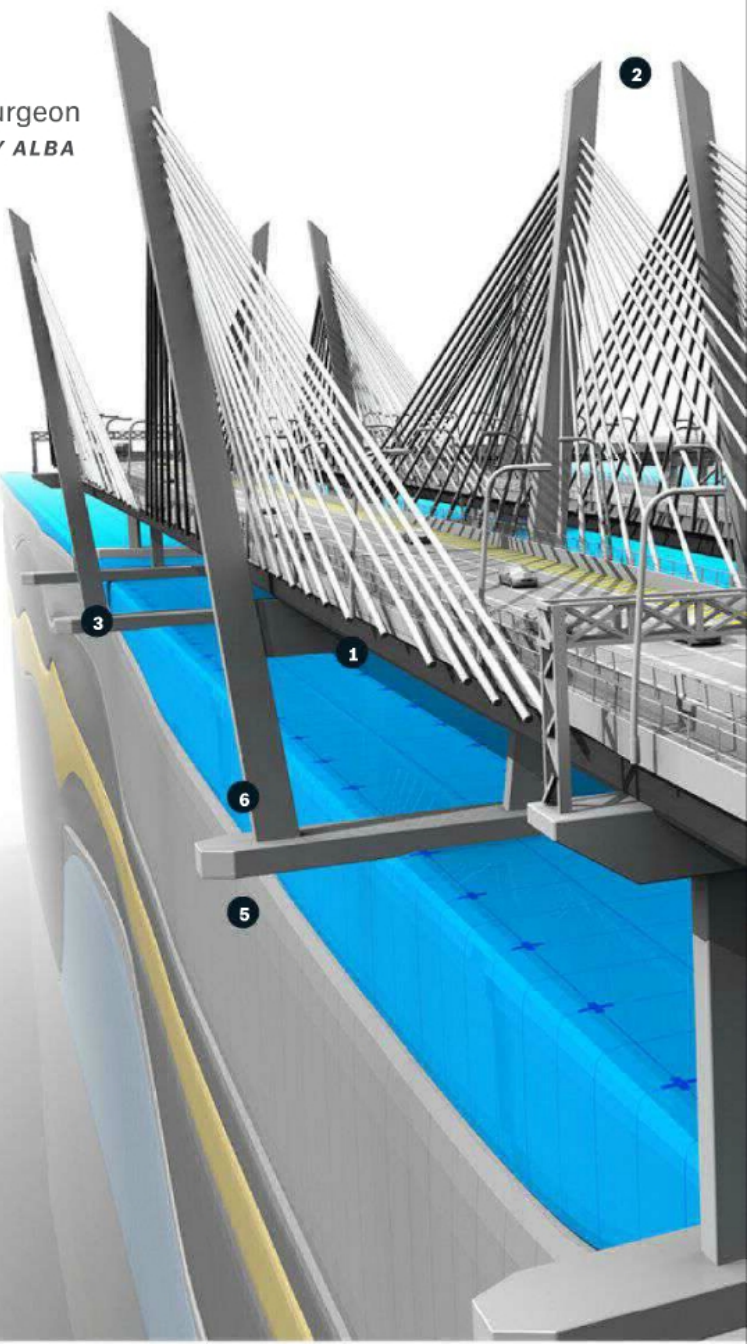
Twenty miles north of New York City, something loud and historic is happening. For the first time in a half-century New York is building a new bridge—a 3.1-mile, $3.9 billion span over the Hudson River. And, this being New York, it's gotta be big. Real big. If you count the 40-foot gap for a possible future commuter rail between the spans (and we do), this will be one of the widest bridges in the world. It will be an accomplishment and a centerpiece—and, best of all, it won't need major repairs for at least 100 years.

The same can't be said of the current bridge, the Tappan Zee. It was lucky to make it this far. Every day 138,000 vehicles drive over it, even though it was meant to handle a maximum of 100,000. Cobbled together on the cheap during a period of material shortages after the Korean War, the existing crossing was built for a 50-year life span. Almost 10 years beyond that, the Federal Highway Administration considers it a "fracture-critical" bridge. Workers had been shoring up the creaky structure since 2007, reinforcing rusting steel supports and patching the crumbling concrete. If the state had decided to keep the bridge, it would have had to spend $3 billion to $4 billion over the next 20 years for maintenance, on top of the $750 million spent on past renovations. The new bridge can't be finished soon enough.

1. The bridge will be made up of two parallel spans with a 40-foot gap for a possible commuter rail—for a combined breadth of 223 feet. When both spans are finished, in 2018, each will have four traffic lanes, two breakdown lanes, and a lane for emergency vehicles and express buses. The luxurious northern span will even have a path to accommodate cyclists and pedestrians.

2. The design of the towers is such that, if the rail line is eventually added between the spans, the towers could be joined at the top to form a pyramid in order to support the additional weight.

3. To buttress the new bridge, crews on barges use enormous, vibrating hammers to drive 964 steel piles—giant, hollow tubes 4 to 6 feet in diameter and up to 360 feet long—into the bottom of the Hudson. The piles on the eastern side of the river are buried 200 to 250 feet deep, through silt and into bedrock. The bedrock on the west side of the river, however, is an unreachable 700 feet below the surface of the water, forcing crews to sink the piles another 100 feet into the riverbed for stability.



AND THE ENORMOUS CRANE THAT HELPED BUILD IT



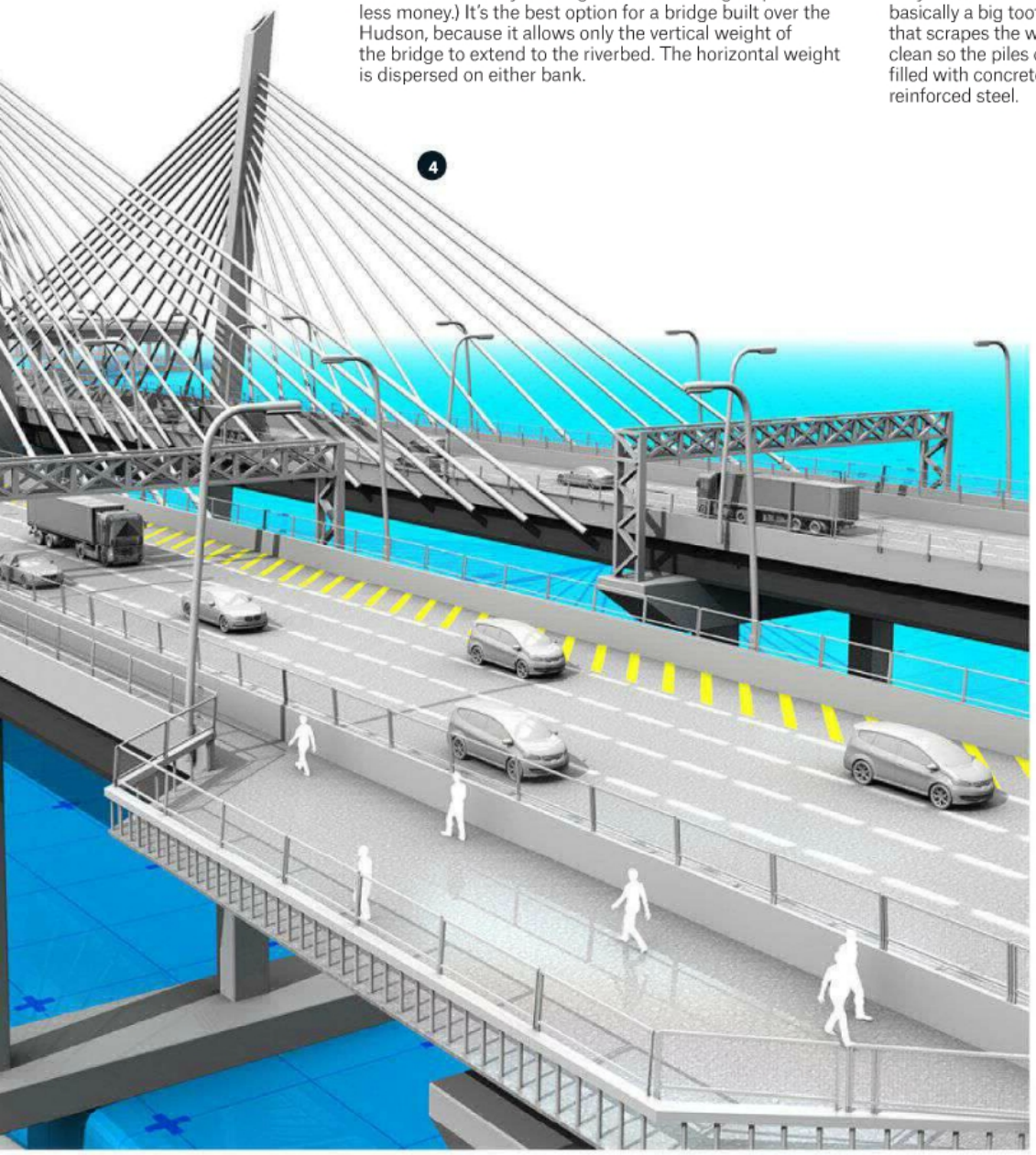
Last winter construction crews called in backup: the Left Coast Lifter, a towering crane that got its name after helping repair the earthquake-mangled eastern span of the San Francisco–Oakland Bay Bridge in 2009.

ILLUSTRATION BY SINELAB

4. To support a thicker, sturdier road deck, the bridge will be cable-stayed; in this case, the weight of the bridge is held up by cables anchored to four towers, each rising 419 feet above the surface of the road. (The once-favored suspension bridge recently gave way to cable-stayed, because cable-stayed bridges can cover longer spans for less money.) It's the best option for a bridge built over the Hudson, because it allows only the vertical weight of the bridge to extend to the riverbed. The horizontal weight is dispersed on either bank.

5. The silt and water that fill each hollow pile as it is plunged into the riverbed are removed by a tool designed specifically for the new bridge—basically a big toothbrush that scrapes the walls clean so the piles can be filled with concrete and reinforced steel.

6. Each tower will be built on clusters of 60 pipes but not before they are tested: An empty barge is placed on top of the pile, then filled with water to create a load of 7 million pounds—more than enough to stand up to a day's traffic.



THE PILE DRIVER

Banging 90-ton piles into bedrock gets noisy, which is bad for two reasons. The first is the neighbors: The state and its contractor are spending $4.2 million to install sound-dampening windows and doors on nearby condo units and homes. Then there are the fish: Resulting sound waves disturb a fish's swim bladder, the gas-filled organ that helps control buoyancy, which could lead to hemorrhaging and death. To limit the effects on the two endangered species that thrive in the Hudson—Atlantic and shortnose sturgeon—every steel pile is wrapped in a "bubble curtain" as it's being pounded in. This aluminum ring slides over the piling like a doughnut on a baseball bat, pumping out air and forming a sleeve of froth that absorbs 10 decibels of sound.



Perched atop a barge longer than a football field, the Lifter sailed 6,000 miles from California, through the Panama Canal and up the East Coast, chaperoned by tugboats. It's one of the largest floating cranes in the world.

With a boom height of 328 feet, arm length of 25 stories, and lift capacity of 1,900 tons (the equivalent of 12 Statues of Liberty), the Lifter will heave large steel girders and prefabricated sections of the road deck into place and help tear down the old Tappan Zee. Using it will shorten the construction schedule by months and help trim production costs by $800 million.